

SANTE-BASSAM — The Project

One project for the whole semester: the health post, what the system is and is not, and the six questions answered before any code

Session 2 of 28 · Friday 11 September 2026 · 10:00 – 11:20 AM

· Prof. Guillaume Sanje Mpacko, Ph.D.

SESSION 2 OF 28

SANTE-BASSAM — The Project

Friday, 11 September 2026 · 10:00 – 11:20 AM · Unit 1 · Foundations and Process

LEARNING OBJECTIVES

- ✓ Describe the health post SANTE-BASSAM serves — its people, its paper, its daily workflow and its non-negotiable constraints — without reference to any software.
- ✓ State what SANTE-BASSAM is and is not: what is in scope this semester, what is excluded, and why the line is drawn there.
- ✓ Answer the six questions from Session 1 for SANTE-BASSAM, with a named person, a number and a dated decision where each demands one.
- ✓ Explain how a decision made in one phase constrains the next, with two concrete examples.
- ✓ Set up the team's working agreement — roles, repository, rhythm, and the fortnightly visit.

FROM SESSION 1

Nine programmes, eight failures, and not one caused by bad code. Six questions that cannot be answered by writing code and get more expensive the longer you wait. Today those six questions stop being a slide and become the first thing you own.

From today, every session opens with fifteen minutes on SANTE-BASSAM, applying what the previous session taught. Today the team itself is the artifact.

WHAT HAPPENS

Practice	The agreement to make now
The team	Six of you, one team. Roles rotate by phase — a Product Owner proxy and a Scrum Master, defined in Session 5. Everyone writes code and everyone tests, including whoever holds a role. One repository on the course GitHub organisation.
Rhythm	Weeks 1–4: no code. Sessions 3–6 choose the process, and weeks 3–4 prepare the repository. Then five two-week sprints, weeks 5–14, each ending with a review.
The post	A fortnightly visit — one review with the nurse, one morning at the desk. The instructor stands in when a visit cannot happen, <i>and the substitution is recorded</i> .
Artifacts	Everything versioned: <code>plan/</code> , <code>srs/</code> , <code>arch/</code> , <code>src/</code> , <code>tests/</code> , <code>handover/</code> . A decision without a dated record in the repository was not made.
Milestones	Six, at the ends of the six delivery phases; each is a tagged commit, reviewed by the instructor and, for two of them, by the nurse.
Missed commitments	Agree the rule <i>today</i> , while nobody has missed anything: what happens when someone does not deliver what they said they would.

Output: `plan/charter.md` in the team repository — members, role rotation, meeting times, the day of the fortnightly visit, and the missed-commitment rule. Committed before you leave the room.

Next session's slot: baseline the three numbers from Q2 and record the architecture decision from Q4 — so today's answers need to be written down, not just said.

1 · Why one project for the whole semester

Session 1 argued that engineering is what happens before the treatment and what makes the treatment accountable afterwards. You cannot learn that from exercises that start fresh each week, because the discipline lives in the **consequences**: the requirement you wrote in week 3 is the test you cannot pass in week 12; the interface you fixed in week 5 is the reason week 9 goes smoothly. So this course uses one system, carried from first conversation to hand-over.

Real enough to be constrained

Real users, real law, real power failures.
Unconstrained problems teach nothing,
because every answer is defensible.

Small enough for six students in fourteen weeks

Alongside four other courses each. Scope
is fixed deliberately, and §3 says where.

Close enough to visit

The single most valuable engineering
activity in this course is sitting beside the
person who will use what you build.

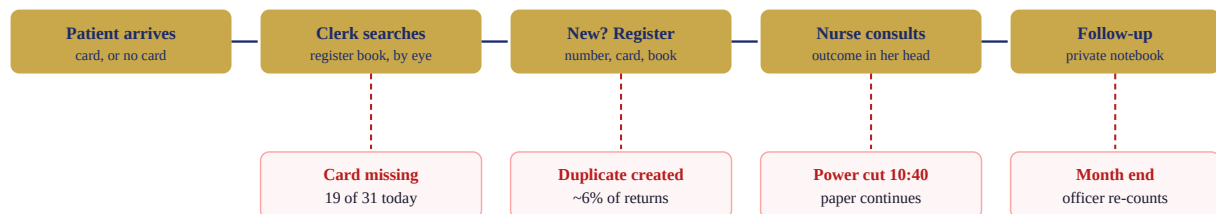
2 · The health post

A community health post in the Grand-Bassam district serves a catchment of roughly 6,000 people across a small town and 23 surrounding villages. It is staffed by a **nurse-in-charge**, who runs consultations and everything else; a **front-desk clerk**, who registers patients, finds

their records and books them in; two **community health workers** who run village outreach several days a week; and a **pharmacy assistant**. A **district health officer** visits monthly and takes away an attendance report that feeds the national statistics. A **management committee** of local notables meets quarterly and can stop anything.

The post's record system is paper, and **it works**. A **register book** lists every patient ever registered, in order, with a number. A **patient card** carries that number and stays with the patient. An **appointment book** lists who is expected on which day. The nurse keeps a **private notebook** of patients she wants to see again, because the register cannot tell her that. At month end the officer **re-counts the register by hand** to fill the Ministry's template.

ONE THURSDAY MORNING AT THE FRONT DESK



Top row: the workflow as the post describes it in an interview.

Bottom row: what one Thursday morning actually contained. The bottom row is the requirement.

Figure 1. The top row is what the post will tell you if you ask. The bottom row is what you see if you sit there for three hours. Session 7 is entirely about that difference.

2.1 Two paper systems — your team chooses one

The register book above is a general description. Real posts keep several registers, and two of them are typical of what software like SANTE-BASSAM replaces. Both are simplified for this course. **In Session 7 the team chooses one**, with the instructor moderating, and writes the choice down as a decision record (ADR-002). Until then, read both with one question in mind: which would you rather specify precisely?

A • The consultation register

Every visit to the post, adults and children, sick or not

Paper tools	A consultation register (one line per visit), a patient card with the patient's number, an appointment book, a monthly activity report filled in by hand.
Typical columns	Line no. · date · patient no. · family and given names · age or birth year · sex · village · new or returning · reason for visit · referred? · next appointment. Diagnosis and treatment columns stay on paper.
Finding a patient	The number on the card; without the card, the clerk scans the book by name.
Follow-up	The nurse's judgement, kept in a private notebook — a rule nobody has written down.
Monthly report	Visits by village, sex, age group, new or returning.
Hardest part	Duplicates, patients without cards, and a follow-up rule that is tacit.

B • The child immunisation register

Children under five and their vaccinations

Paper tools	An immunisation register (one line per child), the child health card kept by the mother, a reminder box of cards filed by the month the next dose is due, a monthly vaccination report.
Typical columns	Line no. · date registered · child's names · sex · date of birth · mother's or guardian's name · village · phone · one date column per dose (BCG, polio, pentavalent 1–3, measles...) · remarks (moved, not seen).
Finding a patient	The mother's name and village — the child's card is often left at home.
Follow-up	Explicit: the next dose has a due date. Children who miss it are traced by the health workers.
Monthly report	Doses given by vaccine and village; how many children dropped out between doses.
Hardest part	Dates and schedule rules, siblings with the same mother, and a guardian on every record.

The worked examples in later sessions use the consultation register (A). If your team chooses B, translate them: the mother's name matters even more, and the follow-up rule becomes a due date.

The first engineering decision has already been made for you

Three facts about the environment will not change and you did not choose them. **Electricity fails roughly once a week**, usually for an hour, sometimes a day. **Connectivity is intermittent** — the town has mobile data most of the time, the villages some of the time, and the post's own connection drops when the power does. **The hardware is one laptop and one Android tablet**, both donated, neither new.

Therefore: anything that needs the network to work *will not be used*, and anything that loses a record when the power fails *will be abandoned within a week* and the post will return to paper. Every later decision in this course is downstream of those two sentences.

3 • What SANTE-BASSAM is — and is not

SANTE-BASSAM is a **patient registry and appointment system**: software that replaces the register book, the card index and the appointment book, runs on the two devices the post already has, keeps working when the power and the network do not, and produces the monthly report without a re-count. That is the whole ambition. It is enough.

In scope this semester

Registry

Register a patient; find a returning patient, including one with no card; detect probable duplicates

Appointments and visits

Book and reschedule; record each visit as attended, missed or rescheduled; the nurse's follow-up list

Outreach

Register and record on the tablet with no network; sync to the desk when connectivity returns

Reporting

The officer's monthly report by village and outcome, in the Ministry template, exportable

Explicitly out of scope

Clinical notes

What the nurse writes about a consultation stays on paper. A clinical record is a different system under different law

Pharmacy stock

The assistant reads visit outcomes; the system does not manage inventory

Time-slot scheduling

The post books a day, not an hour. So does the system

Multiple posts

One post, one server. Records carry a post identifier so a district system *could* merge them later — and that is all

THE OUT-OF-SCOPE LIST IS AS IMPORTANT AS THE IN-SCOPE ONE

In week 10 somebody — a stakeholder, or one of you — will propose adding clinical notes because it seems small. The answer is written here, dated today, and the reason is not effort: a clinical record changes **who may read the data, how long it is kept, and what a defect costs**. **Scope is a requirement**.

4 • The six questions, answered

Q1 — Who is this for, and what will they stop doing?

A person, not a category. The primary user is the front-desk clerk — call her **Madame Koné**, because you will meet her. Today she scans the register book by eye to find returning patients, writes each new patient three times (book, card, appointment book), and re-registers about one returning patient in twenty because she could not find them. She will stop scanning the book, stop writing the third copy, and stop guessing whether a mother’s child is already registered.

The secondary users each give up a habit too: the nurse stops keeping the notebook, the CHWs stop carrying paper sheets back for the clerk to transcribe, the officer stops re-counting. **Each habit is a workaround, and each workaround encodes a requirement.** Session 8 shows you how to read them.

Q2 — What must be true for this to be worth building?

Measure	Today, on paper	Worth building if...	How measured
Time to find a returning patient with no card	~45 s median; fails ~1 in 5	≤ 60 s, fails ≤ 1 in 10	Stopwatch at the desk, 30 patients
Duplicate registrations per 100 returning visits	~6	≤ 1	Count in one month of records
Officer’s hours to produce the monthly report	~3 h re-counting	≤ 10 min, and she submits without re-counting	She tells you; then you watch

If in December the first number is 90 seconds, the project failed even if every test passed. Write these three into the Project Plan and do not soften them later.

Q3 — Which constraints are not negotiable?

Constraint	Why it is fixed	What it forces
Power fails weekly	Infrastructure; not yours to change	Every save is durable <i>before</i> the confirmation appears; the tablet on battery is the desk’s fallback
Network is intermittent	Geography and budget	Offline-first: nothing the clerk or CHW does in a clinic day may depend on the server
One laptop, one tablet, both donated	No budget	Performance targets are measured on <i>these</i> devices; nothing heavier than they can run
Health-record law and the national identifier format	Statute	Identifier format, consent for minors, retention and void-not-delete are domain requirements, never backlog items
The Ministry report template	The officer will not accept anything else	Report columns and their order are fixed; only the counting is ours
French at the desk; more comfortable reading than typing	The people	Interface in French; fields in the order of the paper card; typing minimised — a village list, not free text
Fourteen weeks, six students, four other courses each	The calendar	Scope is fixed by \$3. What flexes is polish, never durability

Q4 — Which decision, if wrong, is most expensive to reverse?

That is your architecture. For SANTE-BASSAM the answer is **where the data lives**, and there are two candidates.

A • Server-centred Shape: one database on a server; laptop and tablet are clients needing a connection.	B • Offline-first chosen, 11 Sep 2026 Shape: each device holds a complete local store; the server is only a sync target; records carry locally generated identifiers.
--	--

Cost if wrong: discovered in week 8 when the CHW cannot register in a village. The whole tablet path is rebuilt — roughly **40 team-days**.

Cost if wrong: if sync proves unreliable the desk still works alone. Rebuild is limited to the sync module — roughly **10 team-days**.

B is chosen not because it is easier — it is harder — but because the cost of being wrong is a quarter of A’s, and because Q3 makes A’s failure near-certain. Session 3 shows how to *test* this decision before committing fourteen weeks to it. The decision record itself goes in the repository today, dated.

Q5 — What is the single point of failure?

If this is unavailable...	...what happens	Mitigation the design must include
The laptop — stolen, broken, dropped	Every record since the last sync is gone; the post cannot register anyone	Sync whenever there is a connection; a weekly export to a USB key the nurse keeps; the tablet can run the desk application
The tablet, in a village	Outreach records since the last sync are gone	Sync before leaving and on return; a paper fallback sheet in the CHW’s bag — which is Q6
The clerk — sick, on leave	Nobody knows how to use the system	The nurse is trained too; a one-page sheet at the desk; no step needs a password only the clerk knows
Your team, after December	Nobody can fix anything	The hand-over sheet, self-explanatory errors, automatic backup — Session 28
The server	Sync stops. Nothing else does	By design, from Q4. The post can run without it for weeks

Read the last row again. Q4 was answered so that the server would not be a single point of failure. That is what it means for one decision to constrain the next — and it is why Q4 comes before Q5 rather than after it.

Q6 — What is the fallback when the clever thing fails?

Denver’s airport had none, and it cost \$560 million. Your registry will run in a building where the power fails weekly and a village where the network does not exist. Q1 to Q5 give you the pattern — a named person, a number, a table with a mitigation column. **Q6 is today’s in-class exercise**, and you answer it to the same standard, in the room, together.

SESSION 2 TAKEAWAY

One project, carried from the first conversation to hand-over, because the discipline lives in the consequences. The environment has already made your first engineering decision: offline-first, or the post goes back to paper. Scope is a requirement, and the out-of-scope list is the half you will need in week 10. And the six questions are not a warm-up exercise — they are the first section of a Project Plan that every later artifact must stay consistent with.

5 · In-class exercise · answer Q6, together

Twenty-five minutes. Three pairs, one context each, then ten minutes to merge into one table the whole team signs. This is the last of the six questions and it is the one Denver got wrong.

Your context

Pair A — the front desk

power fails at 10:40, mid-registration, with eleven people in the queue

Pair B — outreach

the CHW is in a village, the tablet battery dies at 14:00, four registrations unsaved

Pair C — the monthly report

it is the 30th, the officer is here, and sync has not run for nine days

CONTEXT: <desk / outreach / report>

The clever thing : what the software normally does
How it specifically fails: name the failure, not "an error"
Next five minutes : what the PERSON does, step by step
How long it must cover : an hour? a day? a week? give a number
Getting back : how the system recovers, and how paper records taken during the outage get in

Then, as a team: one merged table, committed as `plan/q6-fallbacks.md`, with all six names on it. Together with `plan/charter.md` from this morning and the Q1–Q5 answers above, that is the first section of your Project Plan.

TWO THINGS TO GET RIGHT

“An error occurs” is not a failure mode. “The write is half-finished and the confirmation never appears, so the clerk does not know whether to re-enter” is.

The recovery path is the hard half. Anyone can say “use paper”. The question nobody answers is how the eleven paper registrations from Thursday morning get into the system on Friday without creating eleven duplicates.

No homework this week

Sessions 1 and 2 have no individual assignment — everything due is team work you complete in the room today. Individual homework begins with Session 3, in the Word template you will download from the assignments repository.

Before Tuesday: Sommerville, Chapter 1 §1.1–1.3 and Chapter 2 §2.1.

Where this goes next. You have made one architectural bet (Q4) and written down three numbers you can be judged on (Q2). Tuesday asks the question those answers depend on: **when is a team allowed to find out it was wrong?** Five process models, five different answers, all walked through SANTE-BASSAM.

Key terms

community health post

register book

patient card

appointment book

workaround

catchment

scope, in and out

non-negotiable constraint

offline-first

single point of failure

fallback

milestone

artifact

decision record

Readings and references

- Sommerville, I. *Software Engineering*, 10th ed. — Chapter 1 §1.1–1.3; Chapter 2 §2.1.
- Brooks, F. *The Mythical Man-Month*, chapter 1 — why a program and a programming systems product differ by a factor of nine.

- Pressman, R. & Maxim, B. — Chapter 1, the nature of software.
- World Health Organization, *Analysing Disrupted Health Sectors*, Module 2 — health information at primary level (optional context).